## T1 Rank as a selection rule against the published alternatives - and it is UNDERPOWERED, and its GROUND TRUTH MOVES

Does the metric pick the arm that carries more molecular information? Ground truth: held-out top-CCA on the 40 GENE-SET targets neither arm trained on, confound-residualised block - one of the six blocks banded below.

**n = 6. Exact two-sided binomial: 6/6 -> p = 0.031; 5/6 -> 0.219; 4/6 -> 0.688.**
**THIS DESIGN CAN DETECT A PERFECT RULE AND NOTHING WEAKER. No comparison between two rows of this table is evidence in either direction.**

| metric | D2 s42 | D2 s43 | D2 s44 | D1 s42 | D1 s43 | D1 s44 | D2 | D1 | ALL | exact two-sided *p* |
|---|---|---|---|---|---|---|---|---|---|---|
| **effective_rank_raw** | ✓ | ✗ | unres. | ✓ | ✓ | ✓ | 2/3 | 3/3 | **5/6** | 0.219 |
| **effective_rank_residualised** | ✓ | ✗ | unres. | ✓ | ✓ | ✓ | 2/3 | 3/3 | **5/6** | 0.219 |
| rankme_raw (as published) | ✓ | ✓ | unres. | ✗ | ✓ | ✗ | 3/3 | 1/3 | **4/6** | 0.688 |
| rankme_residualised | ✓ | ✗ | unres. | ✓ | ✓ | ✓ | 2/3 | 3/3 | **5/6** | 0.219 |
| participation_ratio_raw | ✓ | ✓ | unres. | ✗ | ✓ | ✓ | 2/3 | 2/3 | **4/6** | 0.688 |
| participation_ratio_residualised | ✓ | ✓ | unres. | ✗ | ✓ | ✓ | 2/3 | 2/3 | **4/6** | 0.688 |
| stable_rank_raw | ✓ | ✓ | unres. | ✗ | ✓ | ✗ | 2/3 | 1/3 | **3/6** | 1.000 |
| stable_rank_residualised | ✗ | ✓ | unres. | ✗ | ✓ | ✓ | 1/3 | 2/3 | **3/6** | 1.000 |
| alpha_ReQ_raw \|alpha-1\|<br>*authors' released fastssl estimator, not the paper text* | ✓ | ✗ | unres. | ✓ | ✓ | ✗ | 2/3 | 2/3 | **4/6** | 0.688 |
| alpha_ReQ_resid \|alpha-1\|<br>*\|alpha-1\| is OUR operationalisation of their "Goldilocks zone"* | ✓ | ✗ | unres. | ✓ | ✓ | ✗ | 2/3 | 2/3 | **4/6** | 0.688 |
| LiDAR_raw<br>*ADAPTED: q = 2, the two modalities as views* | ✗ | ✗ | unres. | ✓ | ✓ | ✓ | 0/3 | 3/3 | **3/6** | 1.000 |
| LiDAR_residualised<br>*outside the authors' tested q range; ordering invariant over delta 1e-8 to 1e0* | ✗ | ✓ | unres. | ✓ | ✓ | ✓ | 1/3 | 3/3 | **4/6** | 0.688 |

D2 s44 IS MARKED UNRESOLVABLE IN EVERY ROW. F4(a) puts that pair's between-arm gap at 1.4 sampling standard deviations under 40 draws of 80% of patients, so no row of this table can score it. Effective rank's honest D2 record therefore reads: 1 clear hit, 1 clear miss, 1 pair it cannot resolve.
The counts in the D2, D1 and ALL columns are the run's own, which score that cell; they are printed as the run produced them and are NOT corrected here.

LiDAR is ADAPTED - q = 2 with the two modalities as views, licensed by the paper's own footnote 4 but outside the authors' tested q range - and its ordering is invariant across delta from 1e-8 to 1e0. alpha-ReQ follows the authors' released fastssl estimator rather than the paper text, and |alpha - 1| is OUR operationalisation of their "Goldilocks zone", not theirs: all twelve artifacts sit at alpha between 2.6 and 4.8, far outside it. RankMe as published is uncentred on the RAW block, per its own definition; no value in this table is compared against a PUBLISHED RankMe number.

**AND EVERY MARK ABOVE IS SCORED AGAINST ONE COORDINATE SYSTEM OUT OF SIX.**

The ground truth is the held-out channel onto 40 GENE-SET targets. That arm contrast exists on the gene sets and on no other target block on disk (draft 4.6a). Re-scoring the same two arms on each block in turn flips seed 43, and with it every metric row's D2 count. The D1 half cannot be re-scored -- those arms were never scored on any block but the gene sets -- so only the D2 count moves here.

| | gene sets 40 (PUBLISHED) | PBS codes 128 | PCA basis 128 | shuffled PBS 128 | rand control 90 (NEG CTL) | rand dict 128 |
|---|---|---|---|---|---|---|
| arm ordering, seeds 42 / 43 / 44 | **HHH** | **HIH** | **HIH** | **HHH** | **HHH** | **HHH** |
| canonical effective rank (resid.) D2 count | **2/3**<br>(5/6, p=0.219) | **3/3**<br>(6/6, p=0.031) | **3/3**<br>(6/6, p=0.031) | **2/3**<br>(5/6, p=0.219) | **2/3**<br>(5/6, p=0.219) | **2/3**<br>(5/6, p=0.219) |
| RankMe (raw, as published) D2 count | **3/3**<br>(4/6, p=0.688) | **2/3**<br>(3/6, p=1.000) | **2/3**<br>(3/6, p=1.000) | **3/3**<br>(4/6, p=0.688) | **3/3**<br>(4/6, p=0.688) | **3/3**<br>(4/6, p=0.688) |

THE ORDERING BETWEEN THESE TWO ROWS REVERSES ON 2 OF 6 BLOCKS. On the published gene sets RankMe is ahead of canonical effective rank on D2; on the dictionary's own codes and on a plain PCA basis the two swap, and canonical effective rank reaches 6/6 overall - p = 0.031, 'significant' by this table's own bar, produced by nothing but the choice of which coordinate system the exam is written in. No count in this table may be quoted without its target block.